

**COLM
2026**



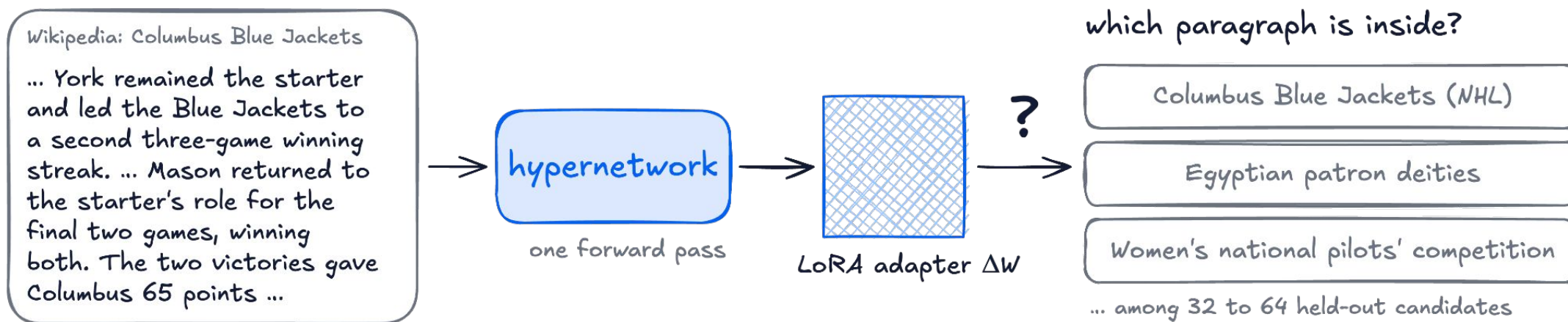
Reading Documents out of Weight Updates: Hypernetwork-Written LoRA Adapters Linearly Encode Their Document

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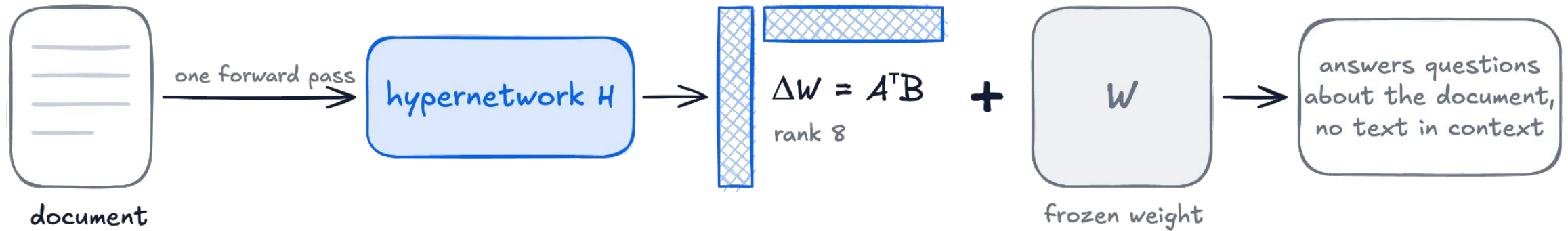
[Context Beyond the Window Workshop, COLM 2026](#)

Motivation



- **Auditing:** was my copyrighted document written into this adapter?
- **Privacy:** can an attacker read a document with private info back out of an adapter?
- **Reconstruction:** works from text embeddings [Morris et al, 2023], still open for weights

Background: Hypernetworks and Doc-to-LoRA

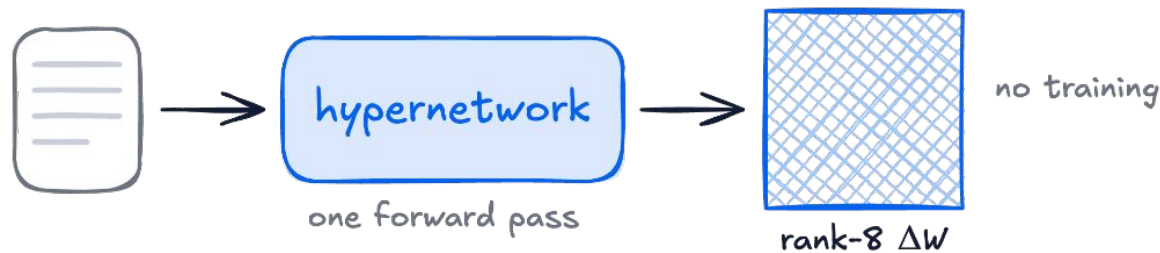


- **Hypernetwork** is a neural network that outputs the weights for another target network [Ha et al., 2017]
- **Doc-to-LoRA** [Charakorn et al., 2026] is trained once (by context distillation), then writes any new document into ΔW in one forward pass. Adds ΔW to the frozen W .
- They **released** hypernetwork checkpoints for three base models

We study 2 ways of writing into weights

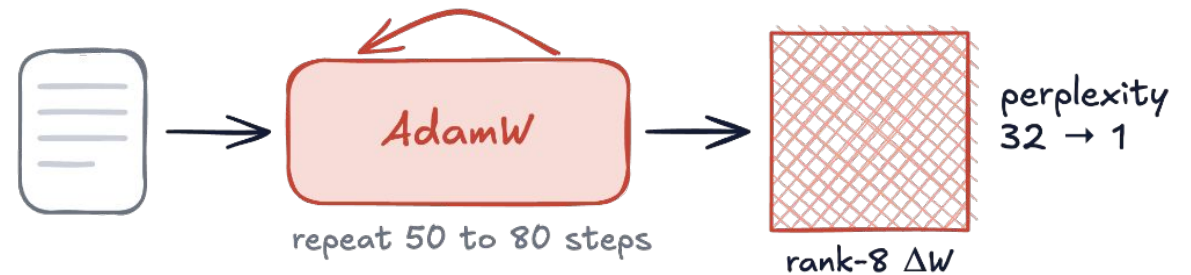
Hypernetwork (Doc-to-LoRA)

- One forward pass per document (instant)
- No training per document



AdamW fine-tuning

- A fresh rank-8 adapter per document
- 50 to 80 training steps



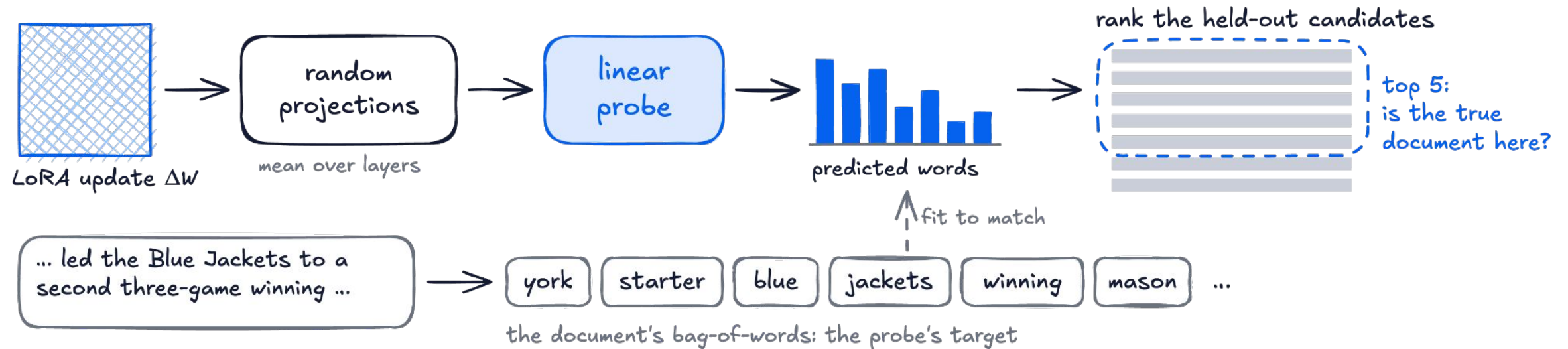
- We use the same factor shape for both, rank 8 on the MLP down_proj
- AdamW adapters learn their document since Gemma's perplexity goes from 32 to 1

Questions we aim to answer

- 1 Can a linear probe identify the document from an adapter's weights?
Does that depend on how the adapter was written?
- 2 What makes a hypernetwork-written update readable if an AdamW update of the same document is not?
- 3 Does identification need white-box access to the weights, or is black-box query access enough?

Identification = given an adapter and N candidate documents, find the one it was written from

Probing the weight update

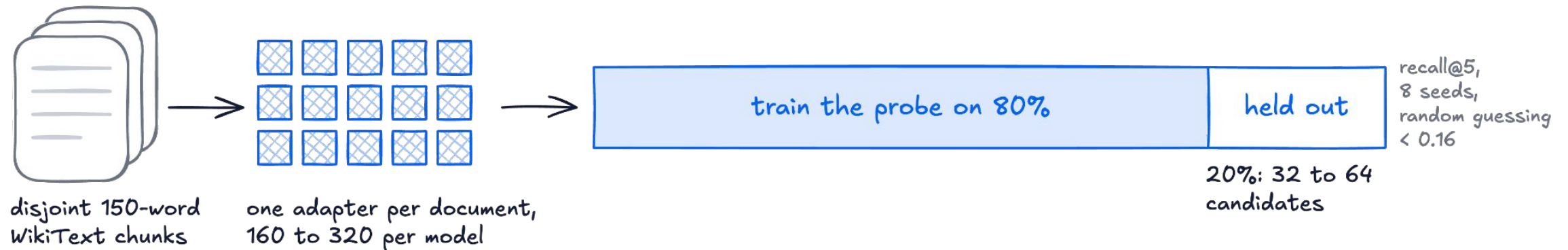


- **Goal** = identify the document among candidates
- **Recall@5** = share of held-out adapters whose true document ranks in the top 5 candidates
- A linear probe asks whether the document is part of the weight update, borrowing from [Alain & Bengio, 2017](#)

Experimental Setup

- **Base models:** Gemma-2-2b-it, Qwen3-4B-Instruct, Mistral-7B-Instruct v0.2
- **Documents:** disjoint 150-word WikiText chunks
- **Adapters:** 160 to 320 per model, one per document
- **Metric:** recall@5 among 32 to 64 held-out candidates

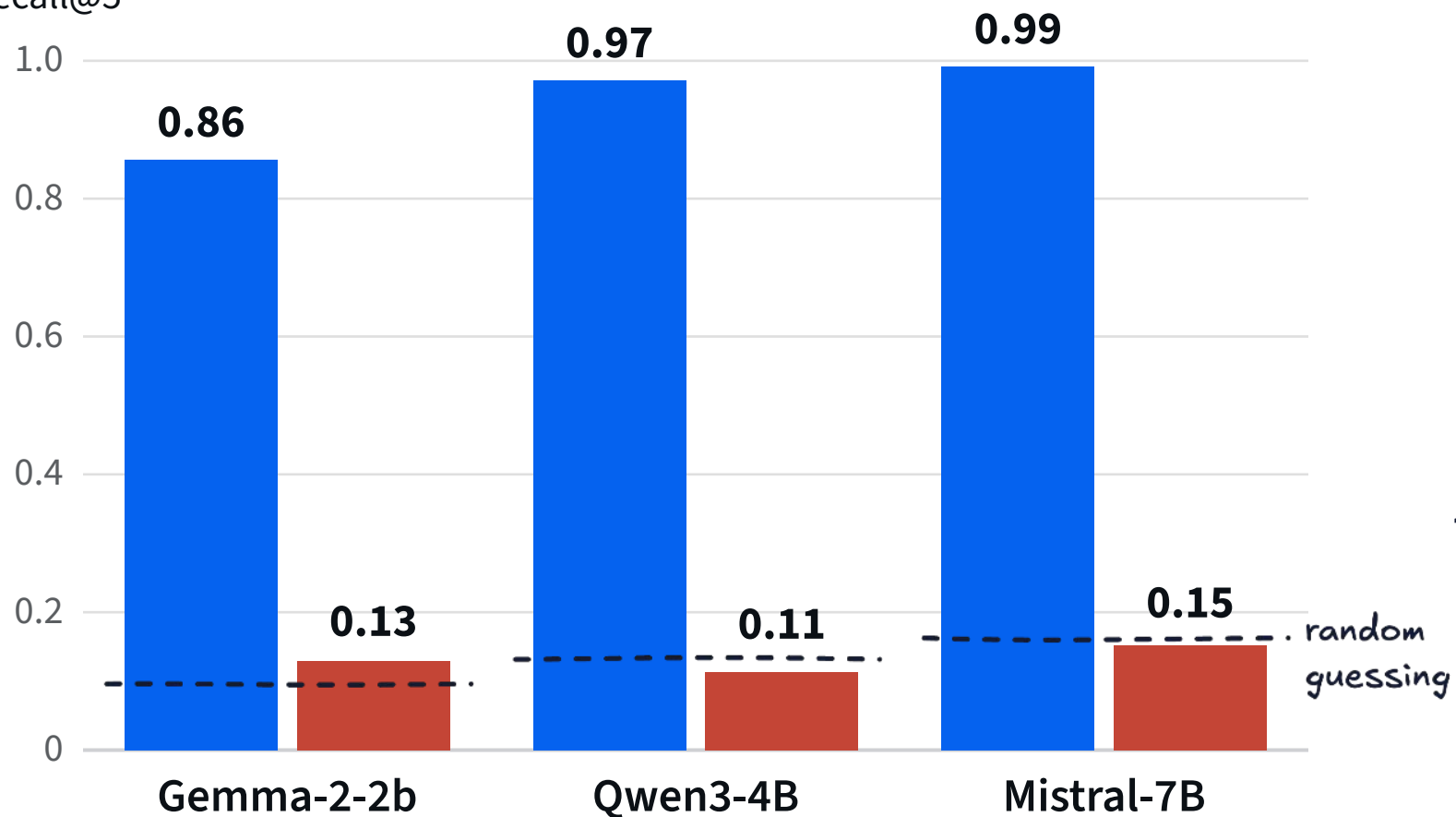
more details in the paper



Results · RQ1

■ hypernetwork ■ AdamW

recall@5



■ **Hypernetwork:**

0.86 to 0.99

**Hypernetwork adapters
can be read, while
AdamW adapters can't**

RQ1 · Recovered words

Held-out document (topic)

Words recovered from ΔW (without text access)

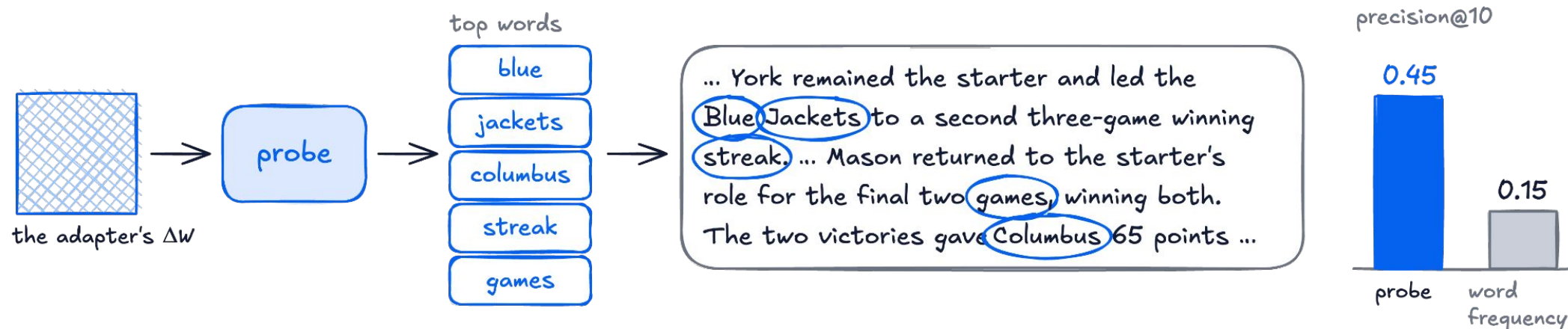
Columbus Blue Jackets (NHL)

blue, jackets, columbus, streak, games

Egyptian patron deities

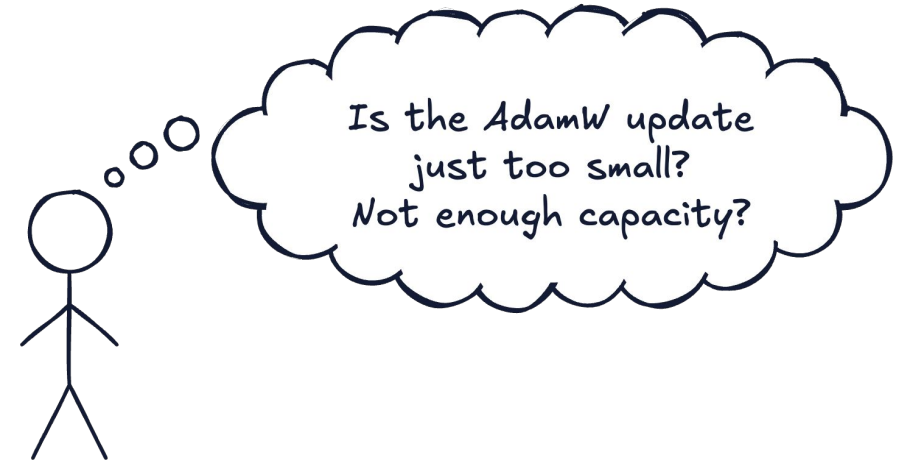
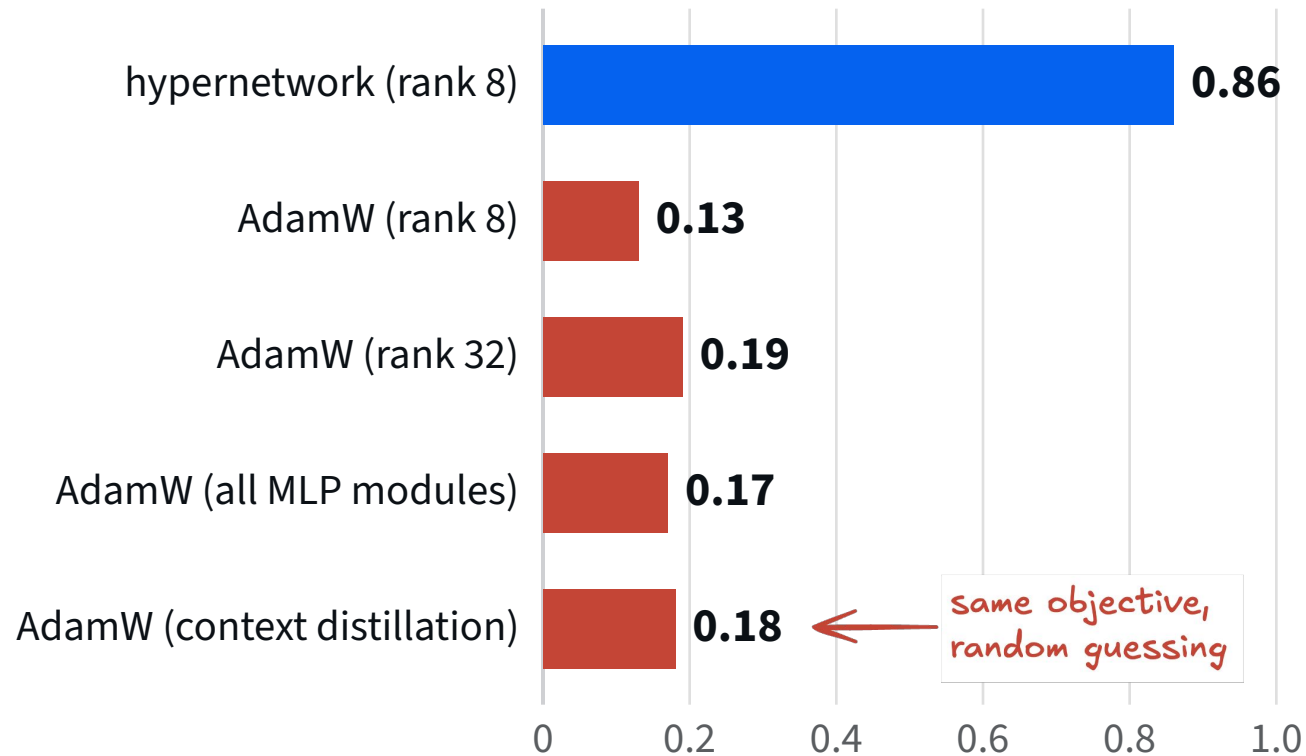
deity, kingdom, amun, name

- **Precision@10:** 0.45 for our probe vs. 0.15 for a word-frequency baseline



RQ1 · Controls

Recall@5 on Gemma-2-2b



- **Update norm:** 6.7 (hyper-network) vs. 5.9 (AdamW)
- **Readability depends on how the update was written**, not on its size, capacity or training objective.

RQ1 · Does it hold for other hypernetworks?

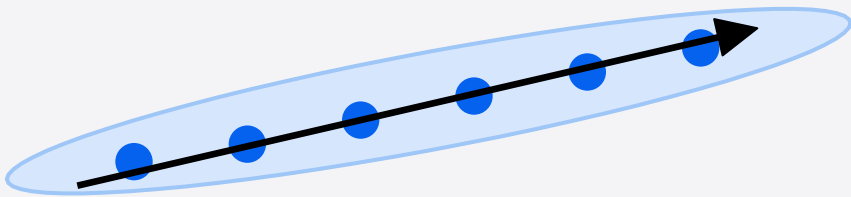
Generator (Gemma-2-2b)	target	recall@5	AdamW
Doc-to-LoRA (released)	MLP down_proj	0.86	0.13
Our hypernetwork (trained from scratch)	MLP down_proj	0.76	0.12
Text-to-LoRA [Charakorn et al., ICML 2025]	attention q_proj	0.85	n/a

All three hypernetworks are decodable, but none of the AdamW adapters we tried are.

Results · RQ2 · Why is the update readable?

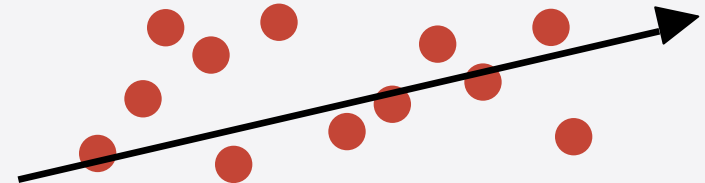
- We believe that a hypernetwork is a smooth map from a document embedding to the update

Hypernetwork



updates lie on a low-dimensional subspace

AdamW



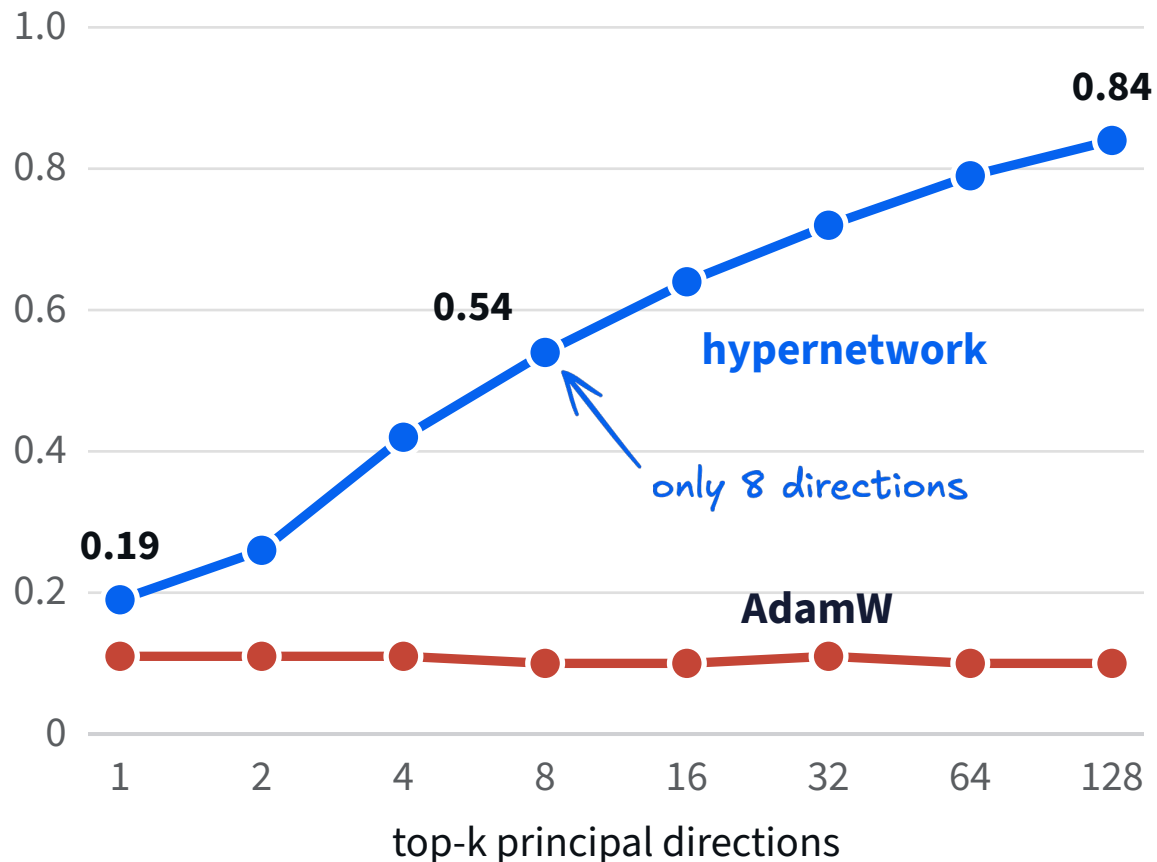
updates fill many directions

- **Linear fit from embedding to update (R^2 on held-out adapters)**

Our hypernetwork (trained from scratch) = 0.97, while AdamW < 0

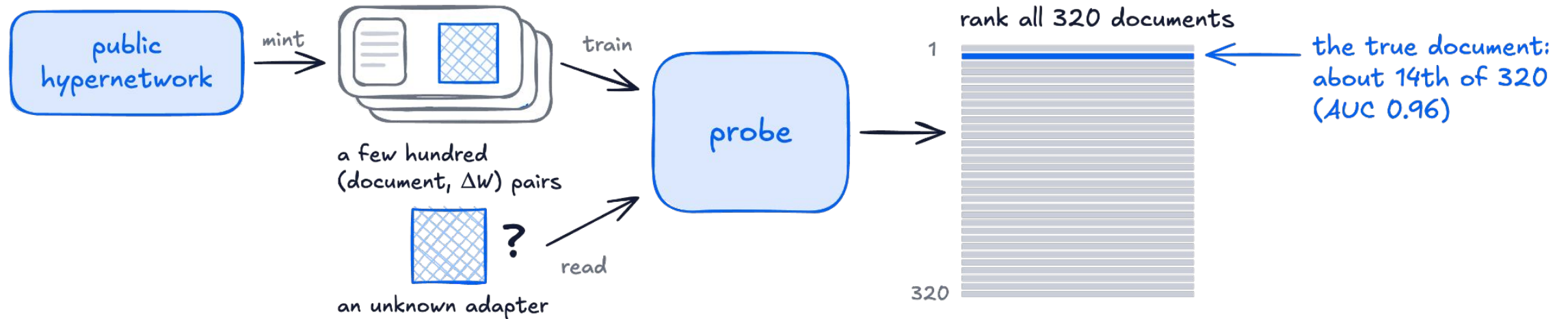
RQ2 · A low-dimensional subspace

Recall@5 using only the top-k principal directions
(Gemma-2-2b)



- **Top 8 directions:** 0.54
(hypernetwork) vs. 0.10
(AdamW)

Results · RQ3 · Real-world application



Audit metric (Gemma-2-2b)	hypernetwork	AdamW
Identification AUC, 320-document pool	0.96	0.50

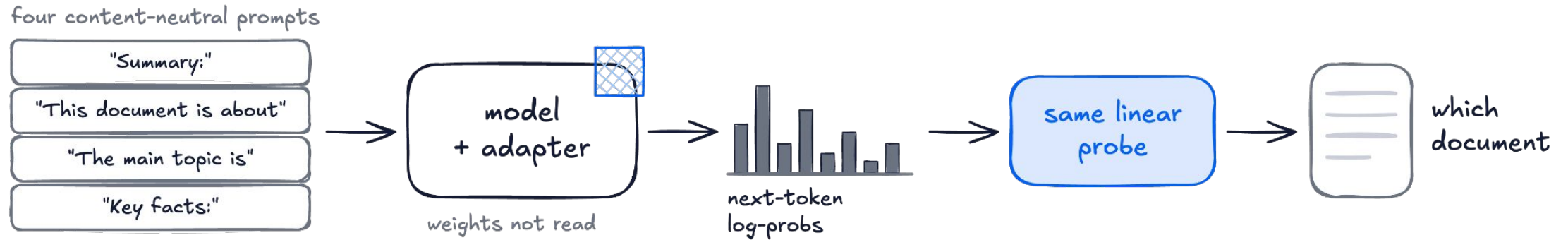
AUC = chance the true document ranks above a random wrong one (0.5 = random guessing). Unlike recall@5, it does not shrink as the pool grows to 320.

RQ3 · Black-box vs. white-box attack



- **Threat model:** the attacker can only query the deployed model with the adapter installed. No weights, no hypernetwork.
- **Attack:** send four content-neutral prompts (e.g., "Summary:", "Key facts:"), and fit the same linear probe to the next-token log-probabilities.

RQ3 · Black-box vs. white-box results

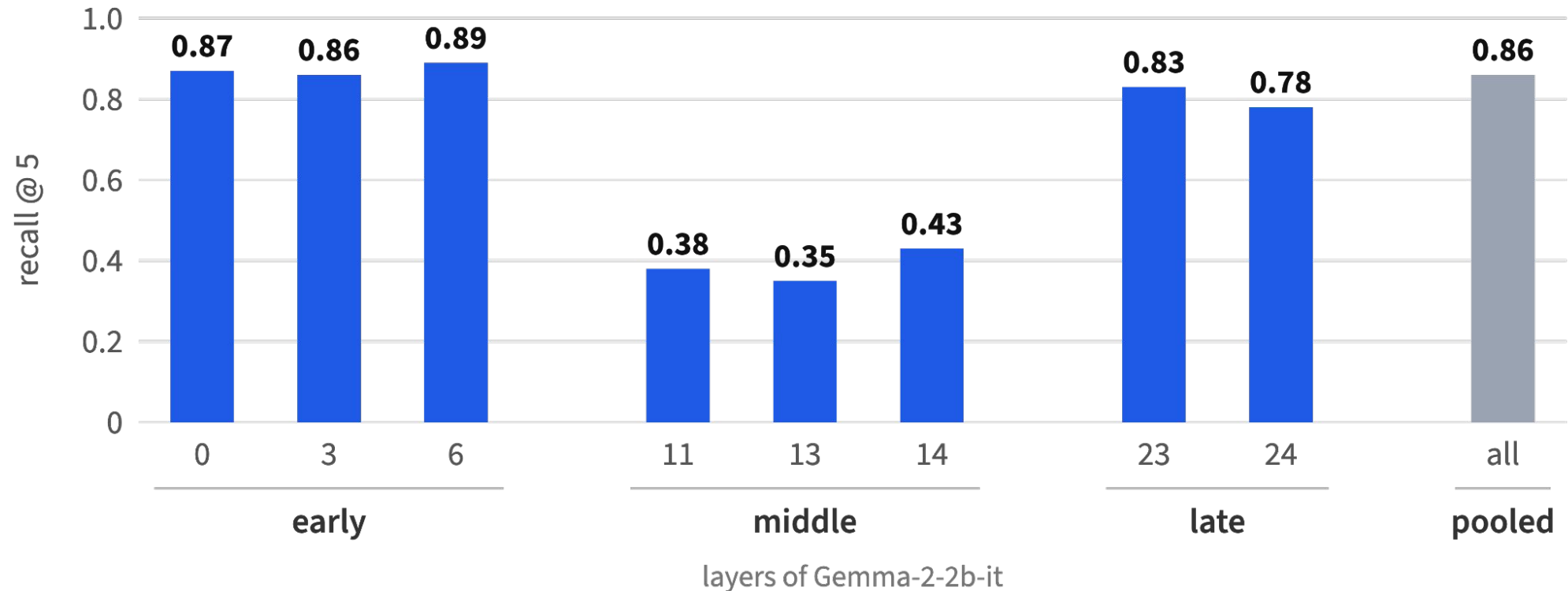


	white-box (read ΔW)	black-box (query outputs)	
recall@5 (random guessing = 0.10)	0.86	0.94	hypernetwork
	0.13	0.59	AdamW

The document is easier to identify from queries than from the weights.

Analysis · Which layers recall the document?

Probing one layer at a time



One early layer recalls the document about as well as all the layers pooled.

Summary

- 1 Can a linear probe identify the document from an adapter's weights?
Does that depend on how the adapter was written?
- 2 What makes a hypernetwork-written update readable if an AdamW update of the same document is not?
- 3 Does identification need white-box access to the weights, or is black-box query access enough?

Summary

- 1** Can a linear probe identify the document from an adapter's weights?
Yes, if a hypernetwork wrote it, on all 3 base models
- 2** What makes a hypernetwork-written update readable if an AdamW...?
For our hypernetwork, we find that the update is close to a linear function of the document embedding.
- 3** Does identification need white-box access, or is black-box enough?
Black-box, ie: querying the model to recall the document, works really well (even for AdamW adapters).

Future directions

- Why did querying the model in the black-box method work so well?
- Reconstruct the document's text from the adapter, not just identify it
- Develop more privacy tests to check for training data identification
- Build an index of adapters, to fetch the one that holds a document
- Effect of release-time changes like lower rank, quantization

Thank you!

TL;DR

Hypernetwork-written LoRA adapters linearly encode their document, so a linear probe on the weight update recalls it while AdamW adapters can't. Black-box queries are surprisingly good at recalling the document either way.

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Paper at [openreview.net](https://openreview.net/pdf?id=Vsuhr8sbG8)
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